

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			{Moves/pushes} <u>mucus</u> (out of the respiratory system) (1) Paralyse cilia/ cilia cannot move / decrease <u>mean</u> length of cilia (1)	2			2		
	(b)	(i)		78 or 79 (mm)	1			1		1
		(ii)		79mm × 15800 = 2 marks If incorrect award 1 mark for 79000/5 79/0.005 79mm = 79000 µm 5µm= 0.005mm 78 mm × 15600 = 2 marks If incorrect award 1 mark for 78000/5 78/0.005 78mm = 78000 µm 5µm = 0.005mm Ecf from (i)		2		2	2	2
	(c)	(i)		Smoking reduces <u>mean</u> length of cilia (1) Greater variation in length of cilia in smokers/ ORA (1)			2	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Increase {sample size / number of people / number of smokers and non-smokers}/ {same/ knowledge of} {age / length of time smoking/ sex/ other health factors/ number of cigarettes smoked per day} IGNORE ref. repeat test			1	1		1
				Question 4	3	2	3	8	2	4